CHAPTER 1

INTRODUCTION TO MICROECONOMICS

CHAPTER ESSENCE AT A GLANCE:
Master fundamental scarcity and economic choice, central problems of an economy, positive versus normative analysis, and production possibility frontiers with opportunity cost.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **The Fundamental Economic Problem:** Human wants are unlimited while resources are strictly scarce and have alternative uses, forcing every society to economize and make optimal allocation choices.

• **Three Central Economic Problems:** Every economy must resolve three core questions: What goods to produce and in what quantities, How to produce them using optimal technology, and For Whom to produce them across households.

• **Economic Organization Frameworks:** Centrally Planned Economies allocate resources through government directives, Market Economies through price signals and consumer sovereignty, and Mixed Economies via joint public-private operations.

• **PPF and Opportunity Cost:** The Production Possibility Frontier illustrates maximum output combinations of two goods given resources and technology; its downward slope reflects scarcity, and concavity reflects increasing Marginal Rate of Transformation.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Marginal Rate of Transformation (MRT)} = \frac{|\Delta Y|}{|\Delta X|} = \text{Marginal Opportunity Cost}$$

Slope of Production Possibility Frontier

$$\text{Central Economic Allocation} \rightarrow \{\text{What to Produce, How to Produce, For Whom to Produce}\}$$

Three Interdependent Economic Problems

$$\text{PPF Concavity Condition} \iff \text{Increasing MOC} \rightarrow \frac{\Delta Y_2}{\Delta X_2} > \frac{\Delta Y_1}{\Delta X_1}$$

Diminishing Factor Suitability Across Goods

$$\text{Opportunity Cost} = \text{Value of Next Best Alternative Foregone}$$

Foundational Valuation of Economic Choice

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse a shift of PPF with a movement along PPF; resource discovery or technological improvement shifts the curve outwards, whereas reallocation moves along the curve.

• **Alert:** Always state that opportunity cost is the value of the next best alternative foregone, not just any alternative.

• **Alert:** Points inside PPF indicate underutilization or inefficient use of resources, while points on PPF indicate full and efficient utilization.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Positive Economics and Normative Economics with one illustrative example each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Positive Economics:** Deals with objective facts and causal mechanisms describing what is without value judgments or ethical opinions. Example: An increase in petrol price leads to a decrease in its demand. **[1 Mark]**

• **Normative Economics:** Deals with ethical assessments, value judgments, and policy ideals describing what ought to be. Example: Government should provide free healthcare to low income families. **[1 Mark]**

2. Explain the three central problems of an economy. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **What to Produce:** Deciding which goods and services to produce and in what quantities (e.g. consumer goods vs capital goods, civil vs war goods) to maximize social utility. **[1 Mark]**

• **How to Produce:** Deciding the technique of production between Labour Intensive Technique (greater employment) and Capital Intensive Technique (greater efficiency and scale). **[1 Mark]**

• **For Whom to Produce:** Deciding the distribution of national product among individuals and factors of production, ensuring functional and personal income equity. **[1 Mark]**

3. Define Production Possibility Frontier (PPF). Why is it concave to the origin? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** PPF is a curve showing various combinations of two goods that an economy can produce with given resources and technology, assuming full and efficient employment of resources. **[1 Mark]**

• **Reason for Concavity:** PPF is concave to the origin due to increasing Marginal Rate of Transformation (MRT) or increasing Marginal Opportunity Cost (MOC). **[1 Mark]**

• **Economic Cause:** Resources are not equally efficient in the production of all goods; transferring resources from good Y to good X forces progressively less suited resources to be transferred, requiring sacrificing increasing units of Y for each additional unit of X. **[1 Mark]**

4. An economy produces two goods: Cotton and Wheat. Given the production possibilities: Combination A (0 Cotton, 15 Wheat), B (1 Cotton, 14 Wheat), C (2 Cotton, 12 Wheat), D (3 Cotton, 9 Wheat), E (4 Cotton, 5 Wheat), F (5 Cotton, 0 Wheat). Calculate the Marginal Opportunity Cost of Cotton at each stage and comment on the shape of PPF. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Formula:** MOC of Cotton = Loss of Wheat divided by Gain of Cotton = Delta Wheat divided by Delta Cotton. **[1 Mark]**

• **Step-by-step Calculation:**
• Combination A to B: (15 - 14) divided by (1 - 0) = 1 unit of Wheat.
• Combination B to C: (14 - 12) divided by (2 - 1) = 2 units of Wheat.
• Combination C to D: (12 - 9) divided by (3 - 2) = 3 units of Wheat. **[1.5 Marks]**

• • Combination D to E: (9 - 5) divided by (4 - 3) = 4 units of Wheat.
• Combination E to F: (5 - 0) divided by (5 - 4) = 5 units of Wheat. **[0.5 Mark]**

• **Economic Conclusion:** Since MOC increases continuously (1, 2, 3, 4, 5), the PPF is strictly downward sloping and concave to the origin. **[1 Mark]**

5. Compare how a Centrally Planned Economy and a Market Economy solve the central problems of an economy. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Centrally Planned Economy:** All major decisions regarding what, how, and for whom to produce are made by a central government planning authority aiming at social welfare rather than private profit. **[1.5 Marks]**

• **Market Economy:** All decisions are coordinated by the decentralized Price Mechanism through forces of demand and supply without government intervention, guided by consumer sovereignty and private profit motive. **[1.5 Marks]**

• **Resolution Comparison:** In planned systems, goods are produced based on national priority plans and distributed administratively; in market systems, only profitable goods demanded by consumers with purchasing power are produced. **[1 Mark]**

6. What does a rightward shift of the Production Possibility Frontier indicate? State two causes for such a shift. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Economic Meaning:** A rightward shift of PPF indicates economic growth and an increase in the production capacity of the economy, allowing higher output of both goods. **[1 Mark]**

• **Two Causes:** (1) Discovery of new natural resources or growth in labour force; (2) Technological advancement improving productive efficiency across sectors. **[1 Mark]**